EXPERIMENT 1

Q.1 Consider following databases and draw ER diagram and convert entities

and relationships to relation table for a given scenario:

COLLEGE DATABASE:

STUDENT (USN, SName, Address, Phone, Gender)

SEMSEC (SSID, Sem, Sec)

CLASS (USN, SSID)

SUBJECT (Subcode, Title, Sem, Credits)

IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

Step 1: From the question, identify main independent objects and choose them as entities → **STUDENT, SEMSEC, SUBJECT** because they represent real-world things with their own attributes.

Step 2: Choose primary keys for each entity (USN for STUDENT, SSID for SEMSEC, Subcode for SUBJECT) because each entity must be uniquely identified.

Step 3: Check tables that only connect entities like **CLASS (USN, SSID)** and treat them as relationships because it only shows which student belongs to which semester-section and cannot exist independently.

Step 4: Treat **IAMARKS** as a relationship/associative entity because marks depend on student, subject and sem-sec, so it connects multiple entities and has its own attributes (Test1, Test2, FinalIA).

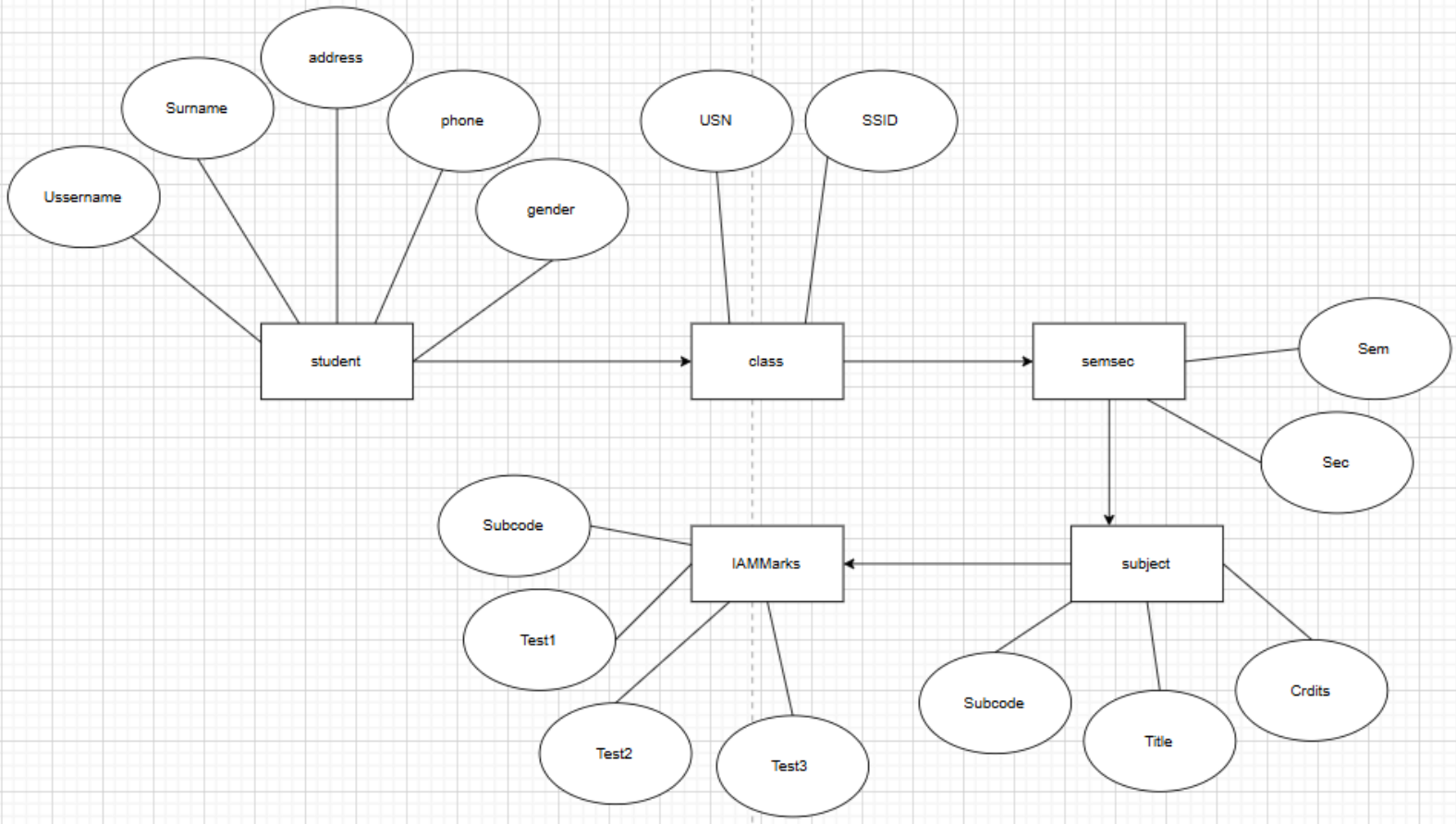
Step 5: Add attributes to each entity and relationship because they describe details like name, phone, credits and marks.

Step 6: Decide cardinality using question context (many students in one sem-sec, one student studies many subjects) because it defines how entities are related.

Step 7: Draw ER diagram using rectangles for entities, diamonds for relationships and ovals for attributes because this is standard ER notation.

EXPERIMENT 1

QUESTION(1)



EXPERIMENT 2

2 Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:

COMPANY DATABASE:

EMPLOYEE (SSN, Name, Address, Sex, Salary, SuperSSN, DNo)

DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate)

DLOCATION (DNo,DLoc)

PROJECT (PNo, PName, PLocation, DNo)

WORKS_ON (SSN, PNo, Hours)

Step 1: Identify main objects from question and choose as entities → **EMPLOYEE, DEPARTMENT, PROJECT** because they exist independently and have their own attributes.

Step 2: Select primary keys (SSN, DNo, PNo) because each entity must have unique identification.

Step 3: Treat **DLOCATION** as weak entity because it depends on department and cannot exist without DNo, so it is linked to DEPARTMENT.

Step 4: Identify relationships from given tables:

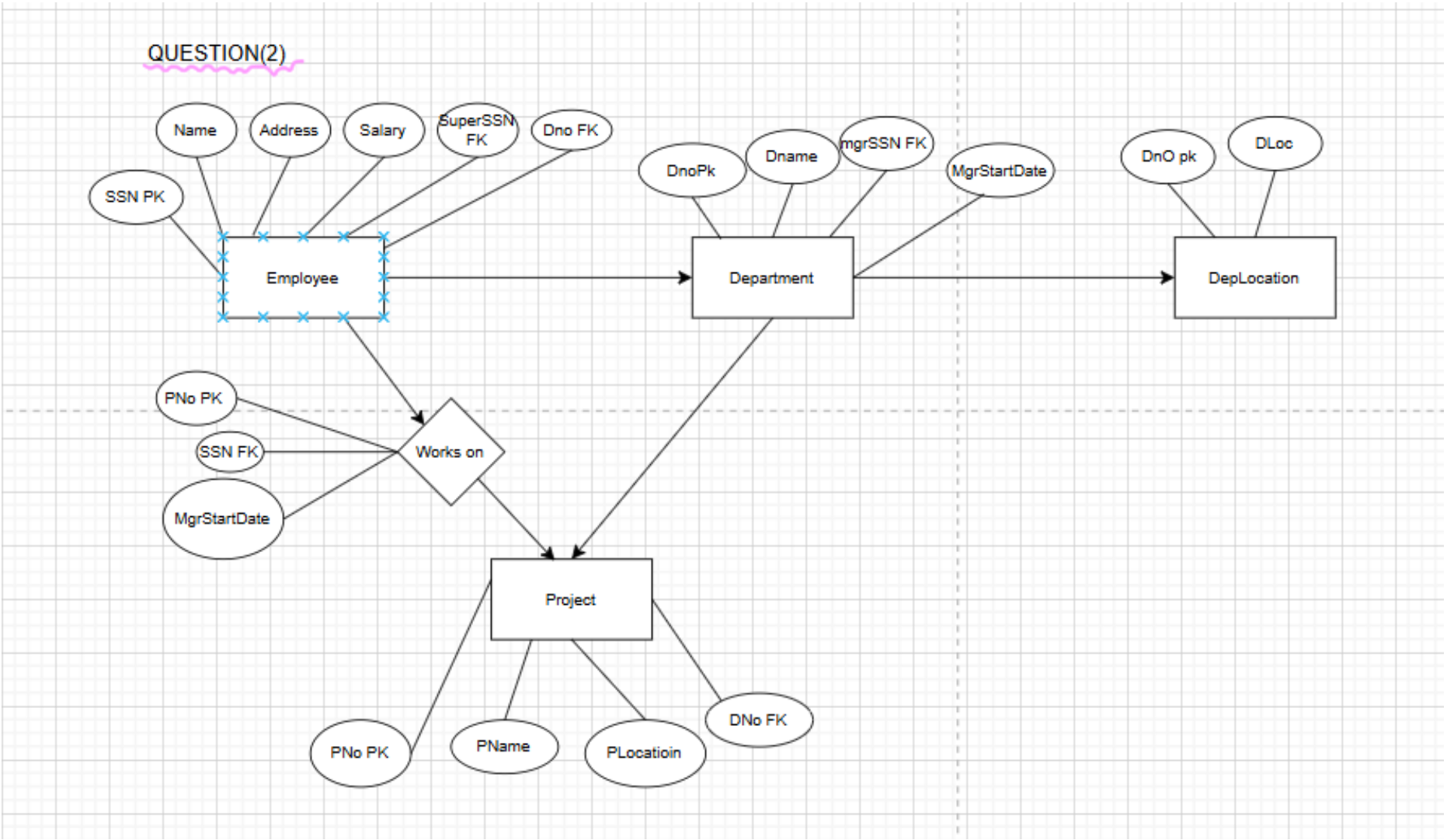
- Employee works in department → relationship
- Employee works on project (WORKS_ON) → many-to-many relationship
- Manager of department → relationship
  because these show interaction between entities.

Step 5: Add attributes to entities and relationships (salary, hours, location) because they describe properties.

Step 6: Decide cardinality from real-world meaning (many employees in one department, many employees work on many projects) because ER diagram must show correct mapping.

Step 7: Draw rectangles for entities, diamonds for relationships and connect with proper cardinality because this follows ER diagram standards used in DBMS.

QUESTION(2)


Name
Address
Salary
SuperSSN FK
Dno FK
SSN PK
Employee
DnoPk
Dname
mgrSSN FK
MgrStartDate
Department
DnO pk
DLoc
DepLocation
PNo PK
SSN FK
MgrStartDate
Works on
Project
PNo PK
PName
PLocatioin
DNo FK